Results-oriented Data Engineering & DBA Leader with 19+ years of expertise in database architecture, ETL/ELT engineering, cloud transformation, performance tuning, HA/DR, and automation across multi-cloud and hybrid environments. Adept at building modern, secure, scalable data platforms that power analytics and business growth. Seeking a leadership role to build and scale robust data and database platforms for high-growth organizations.

CONTACT DETAILS

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+91 9650863867

Ravi Bhatia

CORE COMPETENCIES

- Database Management
- Database Architecture & Engineering
- Migration & Transformation
- Performance Engineering
- High Availability & Disaster Recovery
- Automation & SRE
- Risk Management
- Project Management
- SLA Management
- Team Management
- Cloud Migration
- Security & Compliance
- Budgeting & Cost Control

RAVI BHATIA

DATA ENGINEERING & DATABASE ARCHITECTURE LEADER | ENTERPRISE DATA PLATFORMS | RESILIENCE, PERFORMANCE & SECURITY

PROFILE SUMMARY

- **Data Architecture & Engineering Leader** with **19+ years** across **Oracle, PostgreSQL, MySQL, SQL Server, and cloud-native platforms.**
- Skilled in **Exadata, RAC, ADG, GoldenGate, pglogical, Patroni, Aurora, and InnoDB Cluster design,** and building resilient multi-cloud estates on AWS, Azure, and GCP.
- **Expert in OLTP/HTAP systems, vector databases, and AI-ready semantic layers (pgvector, Milvus, Snowflake Cortex)** for large-scale financial, trading, CPG, logistics, and manufacturing environments.
- Proven track record in **spearheading multi-RDBMS architecture design across** Oracle, PostgreSQL, MySQL, SQL Server, and TiDB to support high-volume OLTP and mission-critical financial workloads.
- **Engineered and managed Oracle Exadata, ZDLRA, and ODA environments** with advanced performance tuning, Smart Scan optimization, and hybrid cluster configurations.
- Experienced in designing and implementing **HA/DR solutions** including clustering, replication, log shipping, and failover to ensure maximum uptime and data resilience.
- Performed deep-dive diagnostics using **AWR/ASH, SQL Monitor, OEM, ExaCLI, OSWatcher** and execution plan analysis to troubleshoot complex performance issues.
- Excellence in optimizing high-volume **PL/SQL, SQL, and ETL workloads** through advanced tuning techniques—partitioning, indexing, rewritten subqueries, materialized views, and parallel query strategies.
- Built **automation frameworks** for Oracle patching, schema deployments, tablespace lifecycle management, and backup validation using **Ansible, Terraform, Python, and Shell.**
- Instrumental in designing end-to-end data ingestion and transformation pipelines using **AWS Glue, DMS, Databricks Delta Lake, Snowflake, dbt, Airflow, and Kafka Connect.**
- **Security & Compliance expertise** spanning **TDE, KMS, Vault, SOX, PCI, GDPR, and Delphix-based masked cloning**.
- Directed **global teams** across **DBA, Cloud, and Data Engineering** to deliver SLAs and large-scale modernization programs.

IMPACTS

- Achieved **40% reduction in query response time**, significantly improving application performance and end-user experience.
- Drove a **30% decrease in infrastructure costs** and operational expenses through database optimization and cloud cost management.
- Minimized **database instance setup time by 70%**, accelerating deployment cycles and improving delivery agility.
- Attained **95% customer satisfaction** in post-implementation feedback through reliable and high-performing database solutions.
- Delivered **50% reduction in manual efforts** by automating repetitive DBA operations, freeing resources for strategic initiatives.

ACHIEVEMENTS

- Engineered robust **database backup and disaster recovery frameworks**, ensured data integrity and minimizing downtime during system failures.
- Orchestrated the **migration of terabytes of PostgreSQL data to Google Cloud** within six months, ensuring zero data loss.
- Advised financial clients in **transitioning from Oracle, DB2 to open-source databases**, achieved substantial cost savings.
- Architected high-availability database solutions, maintained **>99%** uptime and uninterrupted access for critical applications.

PROJECTS EXECUTED

- **Project:** CMC Assignment: Cluster Configurations (Oracle 10g RAC) **Client**: Delhi Sales Tax, NDPL
- **Project:** CMC Assignment: Oracle Upgradation 12c TO 19c (IBM-AIX) **Client:** HAL (Nasik, Bangalore)
- **Project:** CMC Assignment: Cluster Configurations (Oracle 10g RAC) | **Client:** HAL (Nasik, Bangalore)
- **Project:** CMC Assignment: Standby/Dataguard Configuration | **Client:** HAL (Nasik, Bangalore)
- **Project:** TCS Assignment: BI Performance Improvement | **Client:** Agilent Technologies (U.S.A)

TOOLS & TECHNOLOGIES

- Azure Data Lake Storage
- Big Data on Azure
- Big Data on AWS
- PySpark
- Data Structures
- PostgreSQ
- MySQL PostgreSQL (Streaming & Logical Replication, pgvector, Partitioning, FDW, Patroni, PgBouncer)
- Terraform IaC
- Python / Shell Automation
- Prometheus / Grafana
- Data Security & Compliance (GDPR/CCPA)
- Azure Data Factory
- Databricks
- Kafka
- Data Modelling
- Apache Spark Structured Streaming
- Oracle Database
- MySQL (8.0, InnoDB Cluster, Group Replication, ProxySQL, Percona Toolkit)
- Oracle (19c, RAC, ASM, ADG, GoldenGate, Exadata)
- AWS RDS, Aurora, Azure SQL DB

- Fortified data security posture, ensuring compliance and reducing security incidents by **25%.**
- Consolidated multiple database instances, cutting hardware and licensing costs by **20%.**
- Streamlined disaster recovery processes, reducing recovery time from **6 hours to 1 hour.**
- Honored with **Innovation Excellence Award** at Rackspace for pioneering automation frameworks that cut manual effort and cost.
- Recognized for **Outstanding Contribution in Database Management** for excellence in performance, leadership, and innovation.
- Delivered **multi-cloud data architectures** — designed and deployed data lakehouse platforms on AWS (Glue, S3, Iceberg/Hudi) and Azure (Databricks, Delta Lake), enabling zero-downtime ingestion at scale.
- Instituted enterprise-grade governance with **Atlan/OpenMetadata**, automated lineage, schema drift detection, and classification workflows.
- Developed **AWS-native Data Quality Framework** using PySpark + Deequ/Great Expectations, improving data trust and cutting QA turnaround by 40%.
- Optimized cloud performance and cost, reducing AWS Glue job expenses by **30–40%** through intelligent pruning and predicate pushdown.
- Collaborated with **CIOs and senior leaders** (UTZ, Americold) to align architecture decisions with ROI, scalability, and licensing reduction goals.
- Innovated **GenAI-powered analytics modules** *(Smart Segmentation Advisor, Narrative Forecast Explainer, Executive Copilot)* to generate natural language insights from KPI datasets.
- Led integrated **Data Engineering and DBA teams**, unified ETL/ELT development standards, database architecture, HA/DR frameworks, performance tuning, coding and governance practices, automation pipelines, and enterprise-wide data modeling.

WORK EXPERIENCE

May 2025-Present: Leewayhertz (The Hackett Group)
Associate Principal – Database & Data Architect

Key Technologies:
Oracle 19c RAC/ ADG/ GoldenGate/ Exadata/ ZDLRA/ ODA/ CloneDB/ Delphix/ Actifio, PostgreSQL 15–16 (pgvector, Patroni), MySQL 8 (Group Replication), SQL Server 2019 (Always On), Snowflake, Databrick, dbt, Airflow, AWS Glue/Lambda/Step Functions, Azure ADF/Synapse, GCP Dataflow/ BigQuery/ Composer, Terraform, Ansible, Prometheus, Grafana, Python

Key Result Areas:

- **Multi-Cloud & RDBMS Architecture:** Directed enterprise architecture across Oracle, PostgreSQL, MySQL, and SQL Server on AWS, Azure, and GCP, ensuring high availability, scalability, and performance.
- **Database Virtualization & Agile Provisioning:** Engineered hybrid ecosystems with Oracle Exadata, ZDLRA, and ODA; implemented Delphix and Actifio to enable self-service refresh and near-instant cloning, reducing environment provisioning time by 85%.
- **Data Engineering & Analytics Modernization:** Designed and deployed ETL/ELT pipelines using AWS Glue, Databricks Delta Lake, Azure ADF/Synapse, GCP Dataflow + BigQuery; built Airflow orchestration for CDC pipelines and integrated Snowflake + dbt for transformation lineage.
- **AI & Semantic Analytics Enablement:** Integrated pgvector, Milvus, and Pinecone for vector-based semantic retrieval and contextual analytics in AI copilots.
- **Observability, Security & Compliance:** Implemented unified monitoring using Prometheus, Grafana, CloudWatch, Azure Monitor; enforced KMS/TDE encryption, IAM policies, and ensured compliance with SOX, GDPR, CCPA.
- **Practice & Team Leadership:** Headed Database, Cloud Data Engineering, and Performance Architecture practices, mentoring teams, defining standards, and driving adoption of best-in-class solutions across client programs.
- **Process Optimization & Cost Efficiency:** Leveraged CloneDB + Snapshot Copy models, storage-level snapshots, and automation for DevOps/testing, achieving faster turnaround and optimized operational costs.
- **Client & Project Delivery Excellence:** Delivered enterprise-scale data migration, modernization, and analytics transformation projects, enabling high-impact business decisions and measurable operational improvements.

TECHNICAL SKILLS

- **Databases:** Strong in design, modeling, optimization, and security across multi-DBMS environments with focus on integrity and performance.
- **Hadoop:** Proficient in HDFS, MapReduce, and YARN for distributed data processing.
- **PySpark:** Experienced in large-scale processing using DataFrame and RDD APIs.
- **Azure Data Lake Gen2:** Skilled in managing secure, scalable data lakes with Azure analytics tools.
- **IaC (Terraform):** Hands-on with cloud automation and resource provisioning.
- **Automation & SRE:** Terraform, Ansible, Python, GitOps, Observability (Prometheus/ Grafana/ OEM/ CloudWatch)
- **Database Architecture & Engineering:** Oracle (RAC/ADG/Exadata/ZDLRA), PostgreSQL(Patroni/pglogical/pg vector), MySQL (Group Replication/InnoDB Cluster), SQL Server (AlwaysOn)
- **Oracle GoldenGate:** Experienced in configuring and managing real-time replication.
- **Data Replication:** Familiar with AWS DMS and GCP DMS for seamless migrations.
- **Performance Engineering:** AWR/ ASH/ SQL Monitor, OSWatcher, ExaCLI, Query Rewrite, Index/partition design

CERTIFICATIONS

- Oracle Certified Expert (Oracle RAC 10g Administrator), OCP 12c, DBaaS Oracle
- Databricks Certified
- Snowflake Certified
- Oracle Certified Associate
- TiDB Certified

Feb 2024-May 2025: Fidelity International
Technical Consultant- Oracle, Cloud & Data Engineering

Key Technologies: Oracle 12c/19c, Exadata X8, ADG, RAC, ZDLRA, GoldenGate, ODA, Delphix, Ansible, Terraform, Python, Shell, OEM, AWR/ASH, SQL Monitor, Prometheus, Grafana, AWS Glue, Databricks Delta Lake, dbt, Airflow, Snowflake, KMS, IAM, CloudWatch, Azure Monitor, OCI CLI, ServiceNow Integration

Key Result Areas:

- **Hybrid Cloud Database Leadership:** Managed and optimized **Oracle 12c/19c RAC and ADG clusters** supporting fund-valuation, NAV, and order management platforms; integrated on-prem **Exadata X8** with **AWS RDS Oracle** for hybrid DR, cross-region replication, and read-only analytics.
- **Automation & Systems Engineering:**
  - Automated Oracle patching cycles across engineered and non-engineered environments using **Ansible, Python, and REST APIs**, reducing maintenance windows by **60%**.
  - Built self-service provisioning frameworks with **Delphix API + ServiceNow workflows**, enabling on-demand DB cloning and rollback without DBA intervention.
  - Orchestrated **ZDLRA backup automation and Flashback validation** using Terraform and Shell scripts, achieving full compliance with zero data-loss objectives.
- **Performance Optimization & Troubleshooting:**
  - Conducted end-to-end SQL tuning using **AWR/ASH, SQL Monitor, and SQL Plan Baselines**, reducing high-latency batch jobs by **45%**.
  - Implemented proactive bottleneck detection dashboards integrating **Prometheus, Grafana, and OEM** metrics.
  - Optimized PL/SQL modules in order management workflows through refactoring, parallelism, and partition pruning.
- **Data Engineering & Analytics Enablement:**
  - Designed **real-time replication pipelines** from Oracle (via GoldenGate) to Snowflake for analytical and regulatory reporting.
  - Developed incremental ETL pipelines using **AWS Glue and Databricks Delta Lake**, ensuring near-real-time data availability.
  - Implemented **dbt transformations** for schema alignment between Snowflake and Oracle replicas; orchestrated cross-system ETL via **Airflow DAGs** with automated Slack/Teams notifications.
- **Data Security & Compliance:**
  - Configured **Delphix masked data clones** to ensure PII protection in UAT and dev environments.
  - Conducted audits and enforced **GDPR, SOX**, and internal policies using **TDE, KMS, and IAM role segregation**.
  - Automated compliance evidence generation, including RMAN validation, Flashback logs, and DR drill reports.
- **DevOps & Monitoring Integration:**
  - Collaborated with global DevOps teams to integrate **CloudWatch, Azure Monitor, and OEM** events into centralized alerting through SNS and Teams.
  - Developed **Terraform modules** for standardized Oracle provisioning, capacity forecasting, and cost optimization based on historical metrics.

Significant Highlights:

- Automated monthly patching and database refresh cycles, reduced operational time by 60%.
- Delivered AI-ready, analytics-compliant Oracle-to-Snowflake replication pipelines in near real-time.
- Optimized fund-valuation batch processes, shortened execution windows by ~45%.
- Ensured zero data-loss recovery with end-to-end backup automation and validation.
- Enabled developer self-service environment refresh via API orchestration, reclaimed 25% DBA bandwidth for proactive tuning and engineering.

SOFT SKILLS

- Strong Communication
- Analytical Thinking
- Strategic Planner
- Quick decision-making
- Detail-oriented
- Adaptability
- Customer Negotiations
- Multi-tasking
- Innovator

EDUCATION

B.Tech. (Computer Science Engineering)
Kurukshetra University in 2004

Dec 2011-Feb 2024: Tricore Solutions
Lead DBA & Customer Technical Account Manager (Oracle / PostgreSQL / MySQL / SQL Server / Data Platforms) | Rackspace

Key Technologies: Oracle 11g–19c / Exadata / ZDLRA / Delphix / Actifio / NetApp SnapMirror, PostgreSQL 10–15 (Patroni, pglogical, Aurora), MySQL 5.7/8 (Group Replication, PMM), SQL Server 2016–2022 (Always On, Query Store), AWS RDS/EC2/DMS, Azure SQL / Synapse, GCP Cloud SQL / BigQuery, Snowflake, dbt, Airflow, Terraform, Ansible, PowerShell, Prometheus, Grafana

Key Result Areas:

- **Client Engagement & Solutioning:** Acted as Technical Account Manager for enterprise clients; defined HA/DR, replication, and modernization strategies across multi-cloud footprints.
- **Systems Architecture & Virtualization:** Engineered and optimized Oracle Exadata X5/X6 clusters, integrated with ZDLRA for enterprise backup and replication; implemented Delphix and Actifio for rapid database cloning and environment refreshes.
- **Database Administration & Automation:** Managed RAC/ASM, ADG, GoldenGate (Oracle); PostgreSQL streaming/logical replication; MySQL Group Replication; SQL Server Always On; automated provisioning via Terraform, Ansible, and PowerShell.
- **Data Engineering & Modernization:** Built hybrid ingestion pipelines using AWS Glue, Airflow, DMS, feeding Snowflake/Redshift; integrated dbt, Kafka, Tableau, and Power BI for real-time analytics and CDC workflows.
- **Security & Compliance:** Implemented TDE, Database Vault, pgAudit, and unified audit policies, achieving PCI-DSS, HIPAA, and SOX compliance.
- **Leadership & Delivery Management:** Directed multi-practice teams (DBA, Cloud Ops, Data Engineering) managing hundreds of enterprise databases globally; drove operational excellence and on-time project delivery.
- **Automation & Performance Optimization:** Reduced environment provisioning and refresh times by 70%, enabled self-service development/test clones, and optimized SQL/PLSQL workloads for high-volume transaction systems.

Nov 2009-Nov 2011: Mckinsey & Company
Senior Oracle DBA

Key Technologies: Oracle 10g/11g, Exadata V2, CloneDB, RAC, ADG, RMAN, AWR/ASH, OEM, NetBackup

Key Result Areas:

- **Database Administration & High Availability:** Administered Oracle 10g/11g RAC and Data Guard systems supporting global knowledge repositories.
- **Performance Optimization:** Deployed Exadata V2 POC, reducing analytical query latency by 50%.
- **Environment Provisioning & Automation:** Implemented CloneDB and storage snapshot refreshes to accelerate test environment provisioning; automated monitoring scripts for AWR/ASH, session bottlenecks, and tablespace growth.
- **Backup & Disaster Recovery:** Integrated RMAN catalog with NetBackup and managed cross-site replication to ensure data reliability.
- **Collaboration & DevOps Enablement:** Worked with regional DevOps teams to automate patching, failover, and environment refresh workflows.

Aug 2008-Jul 2009: Navisite India Pvt. Ltd.
Senior Oracle DBA

Key Technologies: Oracle 9i/10g, RAC, ASM, Data Guard, RMAN, Delphix (pilot), OEM, MySQL 5.6

Key Result Areas:

- **Database Administration & High Availability:** Managed **Oracle 9i/10g RAC and ASM clusters** across hosted customer environments.
- **Disaster Recovery & Replication:** Configured **Data Guard** and validated quarterly DR cutovers to ensure operational continuity.
- **Migration & Performance Optimization:** Executed cross-platform migrations from file-system to ASM; **optimized redo logs and I/O distribution** for improved performance.
- **Test Environment Modernization:** Piloted **Delphix** for Oracle and MySQL, reducing refresh turnaround and enabling faster development cycles.

PERSONAL DETAILS

Languages Known: English and Hindi
Location: Gurgaon

- **SQL Tuning & Monitoring:** Tuned application queries using **tkprof, Statspack, and EXPLAIN PLAN**; standardized OEM alerts and automated RMAN verification checks.
- **Documentation & Knowledge Transfer:** Created comprehensive **backup, recovery, and migration runbooks** for support teams.

Aug 2007-Aug 2008: HCL COMNET Systems & Services Ltd.
Senior Specialist - Oracle DBA (Project: Bear Stearns)

Key Technologies: Oracle 9i/10g, Exadata (pilot), Data Guard, RMAN, NetBackup, tkprof, UNIX Shell
Key Result Areas:

- Administered Oracle 9i/10g databases supporting risk and trading applications on early Exadata prototypes.
- Configured Data Guard and RMAN with NetBackup integration, ensuring failover readiness for trading floors.
- Optimized SQL performance using Statspack and AWR, achieving millisecond-level trade execution targets.
- Automated snapshot backups and log cleanup through UNIX scripts, reducing manual intervention.
- Collaborated with US trading desks on DR drills and load-balancing tests to ensure operational continuity.
- Prepared SOX-compliant audit reports and supported change control governance.

Oct 2006-Feb 2007: CMC Ltd. (TATA Enterprise)
IT Engineer Oracle DBA

Key Technologies: Oracle 10g, RAC, Data Guard, RMAN, Shell
Key Result Areas:

- Built and optimized Oracle 10g RAC clusters supporting Agilent's supply-chain systems.
- Implemented Data Guard to validate disaster recovery for government client workloads.
- Automated RMAN and snapshot backups along with alert monitoring, enhancing system reliability.
- Supported DR simulations and conducted incident post-mortems to strengthen operational readiness.
- Created comprehensive environment build documentation and RCA templates for support teams.

Feb 2006-Oct 2006: Siel Chemical Ltd.
Oracle DBA

Key Technologies: Oracle 8i/9i, RMAN, PL/SQL, Shell, Statspack
Key Result Areas:

- Administered Oracle 8i/9i databases supporting production, inventory, and accounting systems.
- Implemented RMAN full and incremental backups with daily snapshot exports for plant operations.
- Tuned critical PL/SQL jobs, improving costing report generation time by 50%.
- Migrated and validated legacy schemas from 8i to 9i, ensuring data integrity and minimal downtime.
- Generated monthly KPI and capacity utilization reports for plant leadership.